Prem Ankur Tagdiwala

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I am passionate about Electronics, particularly circuit design and power electronics, and aspire for a career in this field after graduation.

Work Experience

Private Tutoring — Jun 2026 – Present

- Private tutoring to students for GCSE and A Level content in Mathematics and Sciences

CloudyIT — Nov 2022 – Dec 2022

- Worked as part of a team to make an app on MS PowerApps for Haywards Heath Town Council which managed services in the locality

Projects

- LogiSim 8-bit binary adder: made an 8-bit binary adder and multiplexer in LogiSim. Currently recreating them with physical transistors and in Verilog, leading to making an 8-bit CPU.
- Autonomous Underwater Vehicle: in a team of 30+ students to design and build a vehicle that can descend to over 50m in depth. Part of the Power team.
- Hovercraft: worked in a small team, designed the circuit of a miniature RC hovercraft using a Raspberry Pico and MOSFETs to control the fans.
- Radio receiver: made a radio receiver to connect to a phone and receive signals using an adjustable antenna

Technical Skills

- Programming: Python, System Verilog
- Software: LogiSim, Issie, LTSpice
- Electronics: Raspberry Pi, Arduino IDE, Circuit Design, Breadboard, Soldering
- Awards: Northover Prize in Mathematics, UKMT Maths Challenge Gold x6 and BMO participant, Bebras Gold, Senior Physics Challenge Silver, UK Chemistry Olympiad and Cambridge Chemistry Challenge Gold
- Languages: English, Hindi, Gujarati

Education

Imperial College London — London, United Kingdom
MEng Electrical & Electronic Engineering — Sep 2026 – Present

- Kingsbury Scholarship holder

The Royal Latin School — Buckingham, United Kingdom
A Levels — Sep 2019 – August 2026

- OCR MEI Mathematics A*, OCR MEI Further Mathematics A*, AQA Physics A*, OCR A Chemistry A*

Extracurricular

- Integrated Circuits Society: part of the initial team of members founding the society to perform tapeouts of custom-designed integrated circuits
- Integrated Circuits Lecture: researched and delivered a Ted Talk-style lecture to students about integrated circuits - their purpose, logic, and manufacturing process.
- Arduino programming club: member and ambassador for an in-school elective to educate students on programming an Arduino. Helped in setting up and explaining code, and circuit design verification.